Exam 1

ELEE 4210/6210: Linear Systems

Date: August 29, 2025

Time allowed: 50 minutes

Note: Exam is closed book & closed notes. You are only allowed to use a calculator and ONE formula sheet.

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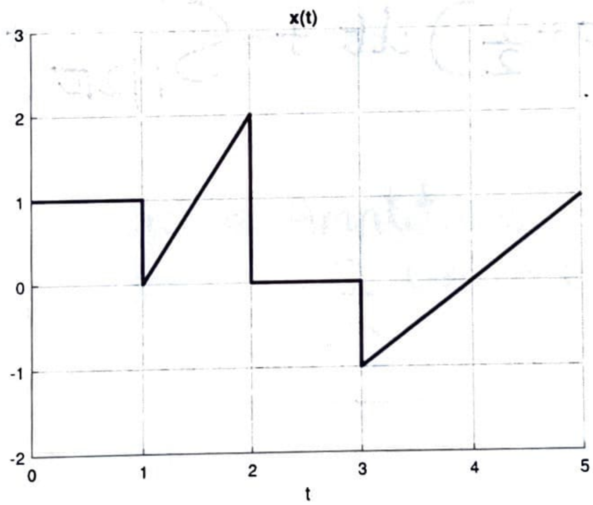
Problem 1 [2+2=4 points]: **(a)** Write an expression for the following waveform in terms of ramp and step functions.



$$x(t) = -u(t-1) + 2r(t-1) - 2r(t-2) - 2u(t-2) - u(t-3) + r(t-3)$$

(b) Calculate the following integral:

$$\int_{-1}^{1} \Big(\delta(1+2t) + \delta(1-2t)\Big) u(t - \frac{1}{2\pi})\, dt.$$

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$$\int_{-1}^{1} (\delta(1+2t) + \delta(1-2t))\, u\left(t - \frac{1}{2\pi}\right) dt$$

As $+\frac{1}{2\pi} \in [-1, +1]$,

$$\rightarrow \cancel{\int_{2\pi}^{1}} \quad \int_{1/2\pi}^{1} (\delta(1+2t) + \delta(1-2t))\, dt$$

$$= \int_{1/2\pi}^{1} \delta(1+2t)\, dt + \int_{1/2\pi}^{1} \delta(1-2t)\, dt$$

$$= \int_{1/2\pi}^{1} \delta(2t+1)\, dt + \int_{1/2\pi}^{1} \delta(-2t+1)\, dt$$

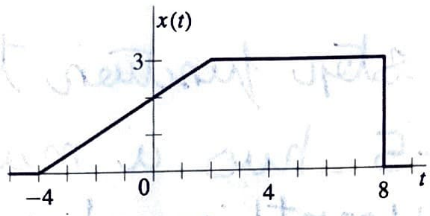
$$= \int_{1/2\pi}^{1} \frac{1}{2} \delta\left(t+\frac{1}{2}\right) dt + \int_{1/2\pi}^{1} +\frac{1}{2} \delta\left(t-\frac{1}{2}\right) dt$$

$$= \frac{1}{2} + \frac{1}{2} = \frac{1}{4}$$

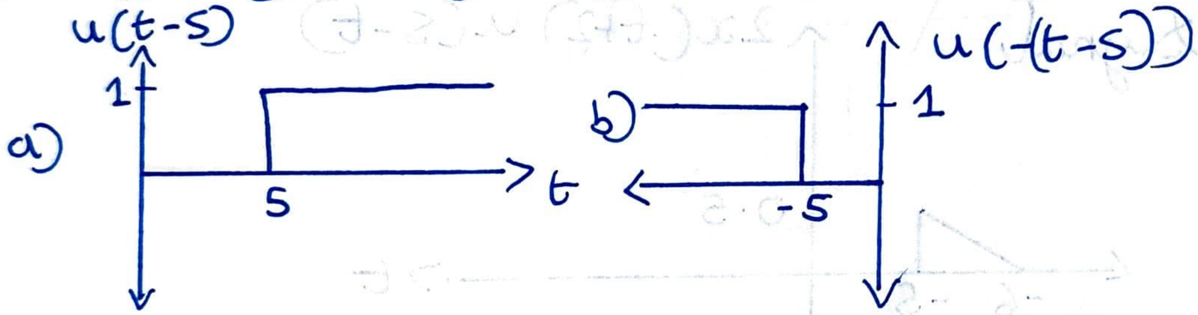
Problem 2 [3 points]: For $x(t)$ shown below, sketch the signal:

$$y(t) = 2x(t+2)u(5-t)$$

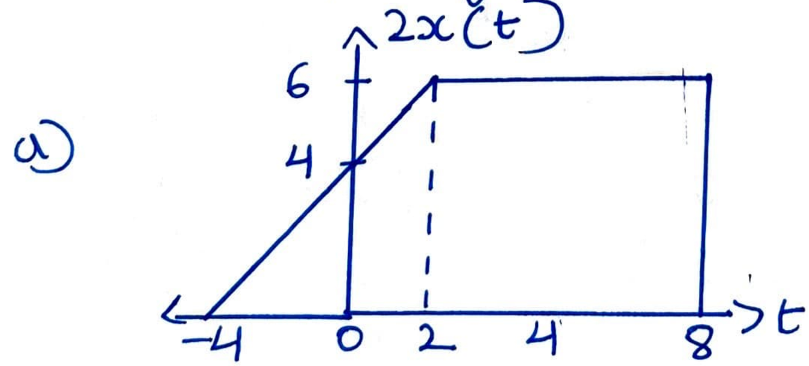
(Note that $u(t)$ is the unit step function)

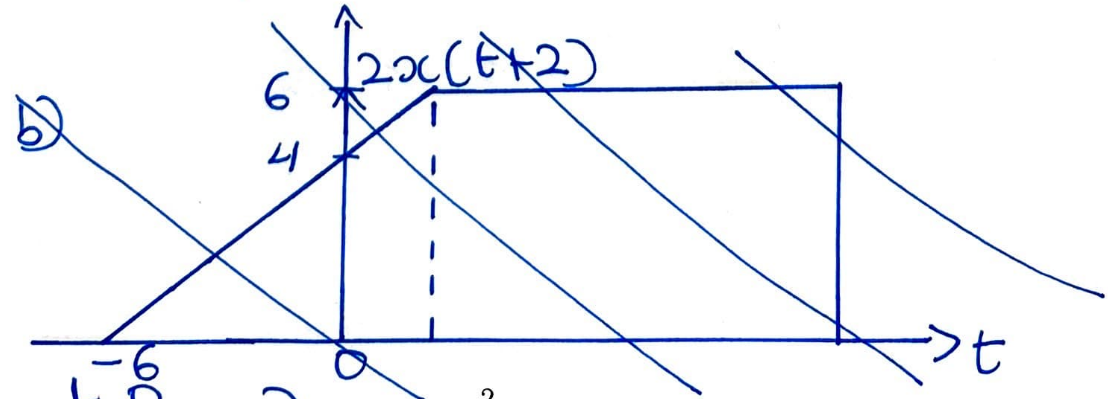


① $u(5-t) = u(-t+5) = u(-(t-5))$



② $2x(t+2) \leftarrow$ Increase Amplitude by 2
Shift by 2 units on x axis left





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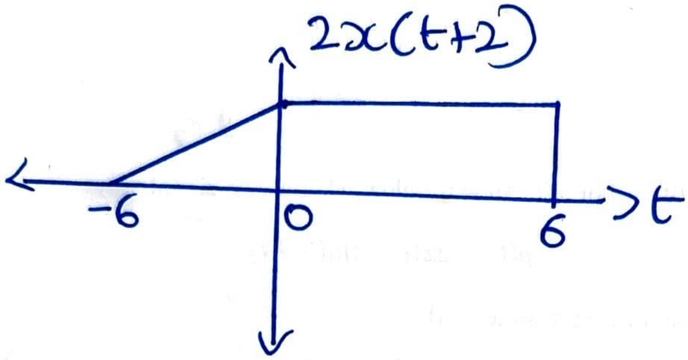
b)

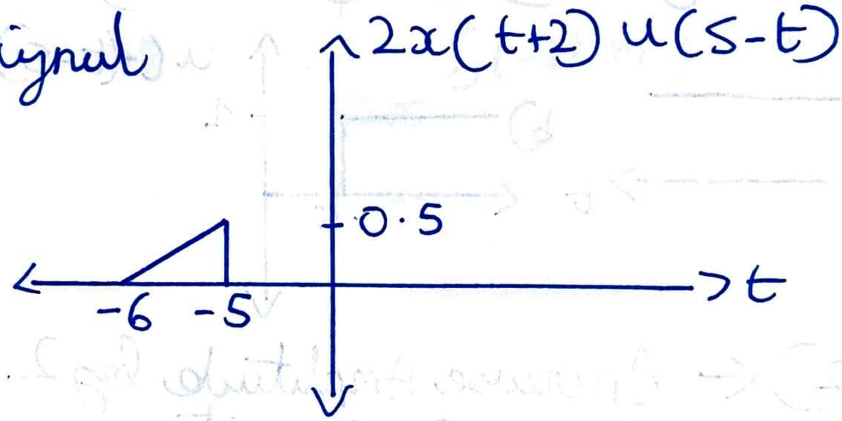


What the unit step function tells us

→ $-\infty < t < -5$ has a magnitude of 1

0 for everything else

∴ Final Signal

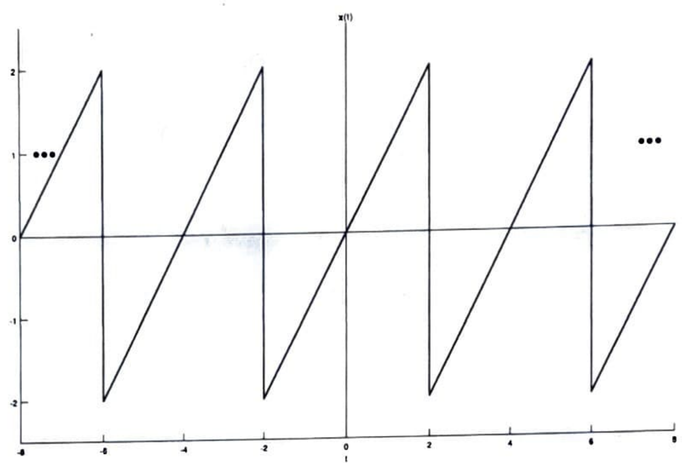


Problem 3 [1.5+1.5+2=5 points]: For the signal $x(t)$ shown below, answer the following questions:

(a) Is $x(t)$ a periodic signal? If so, what is its period in seconds?

(b) Is $x(t)$ even symmetric, odd symmetric, or neither? Why?

(c) Calculate the values of energy and power for the signal $x(t)$. Is $x(t)$ an energy signal, a power signal or neither?



a) This is a periodic signal, It's period is $|-6|-|-2| = 6-2 = 4$ seconds

b) This is neither as this signal isnt a mirror image about the y axis (like for odd ~~even~~ symmetric) nor is it symmetrical about the x and y axis (like for ~~odd~~ even symmetric)

c) The signal is $\cancel{2}r\cancel{(t-1)}\ \sin(1r(t-2)^{-4}u(t-2))$

$E = \infty$

→ This is a power signal as it's periodic.